IS Analytics Competition – Spring 2026

Logistics

Presentations: 4/3 from 1-3pm
Judge's Deliberations: 4/3 from 3-4pm
Awards: 4/3 from 4-5pm

Business Case

The following details will be provided to each team on November 13th:

Coffee Cart Launch Strategy

The University of Utah is considering a **mobile coffee cart** to serve students, faculty, and staff across campus. Before moving forward, administrators want to understand **demand patterns, customer behavior, and operational considerations** that will determine the cart's success.

Your team has been hired as a data analytics consulting group to evaluate a **full year of simulated data** representing coffee sales, campus foot traffic, and temperature. Using Excel and supporting visualizations, deliver insights and recommendations to guide launch decisions.

Proposed Coffee Cart Menu:

Item	Description	Price ($)
Latte	Espresso with steamed milk and light foam.	5.00
Americano	Espresso diluted with hot water.	4.00
Iced Coffee	Brewed coffee served over ice.	4.50
Cold Brew	Slow-steeped cold coffee with smooth flavor.	5.50

Cappuccino	Espresso topped with steamed milk and rich foam.	5.00
Hot Chocolate	Creamy cocoa beverage, topped with whipped cream.	4.25
Iced Tea	Freshly brewed black tea served over ice.	4.50

Time-of-Day Group Definitions

To align sales and traffic data with typical campus activity patterns, each sale or observation is categorized into one of the following time ranges:

- **Morning:** 7:00 AM – 10:59 AM
  - *Peak before and between morning classes.*
- **Midday:** 11:00 AM – 1:59 PM
  - *High lunch-hour activity and foot traffic.*
- **Afternoon:** 2:00 PM – 4:59 PM
  - *Moderate activity as students finish classes.*
- **Evening:** 5:00 PM – 7:59 PM
  - *Reduced campus presence; light sales activity.*
- **Outside Range:** 8:00 PM – 6:59 AM
  - *Non-operating hours; the cart is closed.*

Key Questions to Explore

Your analysis should address the following questions using descriptive statistics, Excel tools, and visualizations:

- **When is coffee demand highest?**
  - Analyze time-of-day, day-of-week, and academic calendar patterns to identify peak sales periods and lulls. Examine how sales volume changes across different hours and seasons.
- **What drinks are most popular?**
  - Identify product trends by item type (e.g., latte, cold brew, hot chocolate) and compare hot versus cold drink preferences across seasonal and temperature conditions.

- **How does weather influence consumption?**
    - Evaluate the relationship between temperature and coffee purchases. Focus on average daily temperature and key weather variables. Note: precipitation has been excluded in this version to simplify analysis.
- **What is the relationship between foot traffic and sales?**
    - Quantify the correlation between EstimatedTraffic and transaction volume. Determine how closely foot traffic levels align with sales activity across time blocks and dates.
- **What operating strategy is most efficient?**
    - Recommend optimal operating hours, menu mix, and service calendar based on demand, weather, and foot traffic patterns. Consider strategies such as reduced summer hours or closure during low-traffic periods.
- **What is the expected range for average daily sales?**
    - Use a random sample of transactions to create a confidence interval for mean daily sales. Interpret the result and explain how sampling variability affects the precision of your estimate.

***Bonus:** Use spreadsheet modeling and what-if analysis to answer the following question:

- What if the coffee cart could only operate 4 hours per day? Which 4-hour window would generate the most revenue?

The accompanying dataset will be provided in Excel format to each team and downloadable in canvas.

Skills Applied

Students will synthesize concepts from throughout the course:

- Descriptive Statistics (mean, median, mode, standard deviation)
- Correlation
- Sampling and Confidence Intervals
- Spreadsheet Modeling and What-If Analysis
- Data Visualization (histograms, scatterplots, box plots, trendlines)

Deliverables

Each team will be required to deliver the following:

1. A written report that summarizes their findings/recommendations based on the business case and accompanying dataset. At a minimum, the report should have the following components:
    a. Executive summary

    b. Methodology, including a description of the tools used
    c. Findings/Recommendations
2. Live 10-minute presentation (using Canva, PowerPoint, etc.) to a panel of judges
    a. A clear framing of the business scenario and objectives
    b. Visuals that communicate key findings and trends
    c. Summary of analytical techniques used in Excel
    d. Final recommendations, with rationale grounded in data

Dataset Description

Below are the sheet names, variables, types, and descriptions/assumptions

Sheet: Estimated Campus Foot Traffic

Purpose: Represents on-campus presence across time blocks and dates to model coffee cart demand.

Variables

- `Date` — *Date*: calendar date for each observation.
- `DayOfWeek` — *Text*: weekday name (Monday–Sunday).
- `DayOfYear` — *Integer*: numeric day of the year (1–366).
- `TimeSlot` — *Category*: period of the day the observation represents.
- `EstimatedTraffic` — *Integer*: estimated number of people near the cart during that time slot.

Sheet: Historic Sales + Weather

Purpose:

Represents simulated transaction-level coffee cart sales data from March 1, 2024 to March 1, 2025. Each row records a single customer order along with the weather conditions at the time of the sale.

Variables

- `order_id` — Integer: unique identifier assigned to each order.
- `Date` — Date: calendar date of the sale.
- `Time` — Time: timestamp of the transaction in 24-hour format (`hh:mm:ss`).
- `hour` — Integer: hour of the day (0–23), extracted from the transaction time.
- `item` — Text: beverage purchased.

    - Possible values: Latte, Americano, Iced Coffee, Cold Brew, Cappuccino, Hot Chocolate, Iced Tea
- `quantity` — Integer: number of items in the order (always 1 in this simulation).
- `price` — Currency: price per item in USD.
- `total_sales` — Currency: total order value, calculated as `quantity × price`.
- `DrinkTemperature` — Category: classifies the drink as Hot or Cold based on item sold.
- `Temperature (F)` — Numeric: outdoor temperature at the time of sale in degrees Fahrenheit.
- `Humidity (%)` — Numeric: relative humidity at the time of sale, as a percentage.
- `Wind Speed (mph)` — Numeric: wind speed at the time of sale, in miles per hour.
- `Cloud Cover (%)` — Numeric: percentage of sky covered by clouds.
- `Weather Type` — Category: general weather condition (e.g., Clear, Clouds).
- `Weather Description` — Text: detailed description of the weather (e.g., scattered clouds, overcast).
- `Season -` Text: categorizes each transaction by time of year based on the date (e.g., Winter, Spring, Summer, Fall).

Operational Rules

- **Weekends:** lower traffic and sales (Friday–Sunday reduction).
- **Prices remain constant** all year.
- **Hours of Operation:** The coffee cart operates between **7:00 AM and 7:59 PM (07:00–19:59)**, covering *Morning*, *Midday*, *Afternoon*, and *Evening* time ranges.
- **Campus Closure Policy:** No sales occur during university holidays, semester breaks, or official closure days as defined by the University of Utah academic calendar.
- **Data Coverage:** Includes all operational days from **March 1, 2024 through March 1, 2025**, excluding closure periods.

Rubric

The following criteria will be used in judging the competition:

Category	Exemplary	Satisfactory	Below Expectations	Points
1. Business Framing & Problem Statement	Clearly introduces the team and business context; problem is well-defined with purpose and measurable success criteria.	Scenario is introduced but lacks detail or focus in the problem definition.	Problem is unclear or missing; little to no context provided.	___ /10
2. Data Exploration & Excel Techniques	Strong use of Excel tools (descriptive stats, PivotTables, charts, correlations); clear interpretation of variables and patterns.	Some analysis is shown but may be incomplete or not fully explained.	Minimal or inaccurate analysis; unclear or ineffective use of Excel.	___ /10
3. Insights & Recommendations	Recommendations are specific, insightful, and clearly grounded in the data; strong business relevance.	Some insights or suggestions are provided but may lack depth or support.	Recommendation s are vague, unsupported, or disconnected from the data.	___ /10
4. Data Visualization & Communication	Visuals are well-designed, relevant, and clearly support the analysis; charts follow best practices (labels, clarity, readability).	Visuals are included but may be unclear, underutilized, or inconsistently formatted.	Visuals are missing, confusing, or poorly executed.	___ /10
5. Delivery & Professionalism	Presentation is organized, confident, well-paced, and engaging; transitions and time use are effective.	Mostly clear delivery with minor timing or structure issues.	Disorganized, unclear, or unprofessional presentation.	___ /10